

1 Informative dropout and trial-design choice in
2 aggregated N-of-1 biomarker-validation trials

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29 1 Abstract

30 **Background.** Aggregated N-of-1 trial designs vary substantially in their robust-
 31 ness to informative dropout, yet the published methodological literature has doc-
 32 umented dropout as a generic problem without adequately engaging the coupling
 33 between the trial-design family (open-label (OL), open-label plus blinded discon-
 34 tinuation (OL+BDC), traditional crossover, hybrid) and the inferential conse-
 35 quences of informative dropout for the biomarker-treatment interaction estimand.
 36 The unaddressed coupling is not a minor sensitivity issue:¹ reports power losses ex-
 37 ceeding 50 percent in some design-by-dropout cells while others are barely affected,
 38 which raises two questions this paper tests. Whether the power consequences of
 39 dropout differ materially across design families at a fixed overall dropout rate, and
 40 whether the most powerful design under no dropout remains the most powerful
 41 under dropout.

42 **Methods.** We formalize the informative-dropout model introduced in¹ (haz-
 43 ard depending on cumulative symptom worsening since baseline), place it within
 44 the classical missing-data taxonomy, and then reproduce and extend Hendrick-
 45 son’s four-design dropout-robustness analysis with wider parameter ranges, sen-
 46 sitivity to dropout-pattern misspecification at the analysis stage, and an explicit
 47 randomization-path decomposition. The simulation employs the existing pmsim-
 48 stats framework with extensions to the dropout-pattern grid, and reporting follows
 49 the² ADEMP framework.

50 **Results.** All four designs were run at a matched total of $N = 70$ partici-
 51 pants, allocated across each design’s randomization paths, at 500 replicates per
 52 cell over a 16-cell design-by-dropout grid with complete convergence. Without
 53 dropout, power to detect the biomarker-treatment interaction was 0.968 (hybrid),
 54 0.950 (OL+BDC), 0.854 (traditional crossover) and 0.088 (OL), and that order-
 55 ing was unchanged under all three dropout conditions. The hybrid advantage
 56 over OL+BDC was narrow and reached significance in only one of the four con-
 57 ditions: 1.8 percentage points without dropout ($z = 1.44$), 5.4 under 25 percent
 58 uninformative dropout ($z = 3.59$), 2.4 under 25 percent worsening-driven dropout
 59 ($z = 1.55$) and 1.6 under 40 percent worsening-driven dropout ($z = 0.85$). The
 60 OL design is not a viable comparator for this estimand rather than a robust one:
 61 with the exposure indicator identically one across its single all-on-drug path, the
 62 interaction term is aliased with the biomarker main effect and the design carries
 63 essentially no information. Under 40 percent worsening-driven dropout the tradi-
 64 tional crossover lost 8.6 percentage points of power against 5.8 for the hybrid and
 65 5.6 for OL+BDC, making the crossover the only design clearly separated on ro-
 66 bustness. Dropout rate dominated dropout mechanism: at a fixed 25 percent rate,
 67 worsening-driven and uninformative dropout differed by at most 1.8 percentage
 68 points in any design, none of them distinguishable at this replicate count. Infor-
 69 mative dropout cost precision rather than accuracy. Within each powered design

the mean interaction estimate moved by less than 0.08 across dropout conditions on a coefficient near -2.2 to -2.5 , under a tenth of its within-cell standard deviation, while the mean model-based standard error rose by 13 to 14 percent from no dropout to 40 percent biased dropout. We found no evidence that selective retention of treatment-non-responders inflates the apparent interaction.

Conclusions. Designs should be compared at a matched total sample size. An earlier unmatched version of this analysis gave the four-path hybrid design four times the sample of the open-label design and, with hybrid power then sitting against the ceiling, produced the appealing but spurious conclusion that the most powerful design is also the most dropout-robust; at matched N the hybrid design's power loss is indistinguishable from OL+BDC's. On these operating characteristics the hybrid and OL+BDC designs are interchangeable, so the choice between them should turn on participant burden and feasibility, while the traditional crossover is the design to avoid when heavy worsening-driven dropout is anticipated. Trial designers should weigh anticipated dropout rate first and mechanism second, since mechanism effects were not resolvable here; separating them would need approximately 2{,}000 replicates per cell. We provide a design-by-dropout recommendation table on this basis.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[1]

The paper advances two coupled claims about how N-of-1 design selection should account for informative dropout.

- The four design families (OL, OL+BDC, traditional crossover, hybrid) differ in dropout robustness in ways the published literature has not adequately characterized.
- Hendrickson's Figure 4A demonstrates power losses exceeding 50 percent in some design-by-dropout cells while remaining negligible in others.
- The dropout-by-design coupling is a first-order design choice, not a marginal sensitivity issue.

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We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[2]

The first claim is that design families differ in dropout ro-

109 **bustness in ways the published literature has not adequately**
110 **characterized.**

- 111 • Power loss from a realistic dropout pattern can exceed 50 percent
112 in some design-by-dropout cells.
- 113 • Design rankings reverse depending on the dropout pattern.
- 114 • This is a first-order design choice, not a marginal sensitivity issue.

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117 The first is that the four candidate trial-design families (open-label,
118 open-label plus blinded discontinuation, traditional crossover, and the
119 hybrid N-of-1 design proposed by¹) differ in their robustness to infor-
120 mative dropout in ways that the published methodological literature
121 has not adequately characterized. Hendrickson’s Figure 4A heatmap
122 demonstrates that the absolute power loss from a realistic dropout
123 pattern can exceed 50 percent in some design-by-dropout cells while
124 remaining negligible in others, and the design rankings reverse depend-
125 ing on the dropout pattern. This is not a marginal sensitivity issue
126 but a first-order design choice that the field has not, to the best of our
127 knowledge, foregrounded.

128 [3]

129 **The second claim is that the dropout mechanism, not merely**
130 **the rate, is the methodologically relevant object for design**
131 **selection.**

- 132 • Balanced, worsening-driven, and improvement-driven mecha-
133 nisms interact with design families in qualitatively different
134 ways.
- 135 • Trial designers should evaluate designs against the anticipated
136 dropout mechanism rather than a generic rate.
- 137 • The mechanism is the primary object; the rate is relatively super-
138 ficial.

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141 The second claim is that the dropout-pattern mechanism (balanced,
142 symptom-worsening-driven, symptom-improvement-driven) interacts
143 with the design family in qualitatively different ways than the overall
144 dropout rate alone, and that trial designers should therefore evalu-
145 ate proposed N-of-1 designs against the anticipated dropout-pattern
146 mechanism rather than against a generic dropout rate. That is, the
147 mechanism is the methodologically relevant object; the rate is rela-
148 tively superficial.

149 [4]

The paper imports the mature missing-data taxonomy into the N-of-1 design-comparative context for the first time.

- The missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) taxonomy and the mixed-effects model for repeated measures (MMRM) have been worked through for parallel-group RCTs but not for N-of-1 designs comparatively.
- ICH E9 (R1) formalizes estimand-aware handling of intercurrent events including dropout.
- The present paper undertakes the import of this machinery into the N-of-1 design-comparative setting.

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The case is, in the broadest framing, a special application of the missing-data taxonomy of Little and Rubin³ to the N-of-1 design class. The taxonomy is mature: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) define a ladder of assumptions about the missingness mechanism, and each layer imposes correspondingly stronger inferential conditions^{4,5}. The classical clinical-trial literature has worked through the implications for parallel-group randomized trials⁶⁻⁸ and has converged on the mixed-effects model for repeated measures (MMRM) as the default analytic strategy when dropout is plausibly MAR. The ICH E9 (R1) addendum⁹ formalizes the estimand-aware framework that handles intercurrent events including dropout¹⁰. Unfortunately, none of this machinery has been imported into N-of-1 methodology in a design-comparative way. The present paper undertakes that import.

2.2 Why the question matters

[5]

Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class.

- Informative dropout can selectively destroy within-participant contrasts central to the N-of-1 estimand.
- Patient self-selection is correlated with latent biomarker-response components that are partly signal, not nuisance.
- The four canonical designs differ in their phase-specific vulnerability windows.

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Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class. We shall take each in turn.

[6]

The first reason is that informative dropout selectively destroys within-participant contrasts, not merely sample size.

- Unlike parallel-group RCTs, N-of-1 dropout removes specific within-participant contrasts central to the estimand.
- A participant lost before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation.
- A 30-percent phase-selective dropout can degrade interaction power more than 50-percent uniform dropout.

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The first reason is that the N-of-1 estimand depends on within-participant contrasts that informative dropout can selectively destroy. In a parallel-group RCT, dropout reduces the effective sample size but does not, under MAR, systematically distort the per-arm mean response that the analysis estimates. In an aggregated N-of-1 trial, dropout removes specific within-participant contrasts: a participant who drops out after the open-label phase but before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation that the design is constructed to deliver. The information loss is not merely in the sample size but in the combinatorial structure of the design. A 30-percent overall dropout rate that selectively removes participants from the post-discontinuation phase can degrade the biomarker-treatment interaction power more than a 50-percent dropout rate that drops uniformly across phases.

[7]

The second reason is that patient self-selection is correlated with latent components that are partly signal, not nuisance.

- High-BR patients tend to remain on drug; high-PB patients may drop at discontinuation; negative-TV patients may drop from worsening.
- Each mechanism produces a different dropout pattern that interacts differently with the design family.
- The worsening- and improvement-driven patterns operationalize the responder-versus-non-responder distinction the biomarker literature is validating.

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The second reason is that patient self-selection into and out of the trial is correlated with the latent BR-PB-TV components in ways that are not nuisance but partly signal. Patients with strong

pharmacological response (high BR) are likely to remain on drug and therefore to remain in the trial; patients with strong placebo response (high PB) may drop out at the blinded-discontinuation phase when their symptoms return on the silently-switched placebo, mistaking the return for trial failure; patients with adverse natural-history trajectories (negative TV) may drop out from worsening regardless of treatment. Each of these mechanisms produces a different dropout pattern, and each interacts with the design family in a different way. The 'symptom-worsening-driven' and 'symptom-improvement-driven' patterns of¹ are operationalizations of the responder-versus-non-responder distinction that the predictive-biomarker literature is endeavoring to validate; the dropout pattern itself is therefore not incidental.

[8]

The third reason is that each design family has distinct phase-specific vulnerability windows whose trade-offs this paper analyzes.

- OL is most dropout-robust but cannot identify the BR -versus- PB split required for the predictive-biomarker question.
- OL+BDC introduces a discontinuation-phase window where responders may drop from symptom return.
- The hybrid design has the largest vulnerability surface; the trade-off between identifiability and dropout vulnerability is the central design choice.

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The third reason is that the four canonical designs differ in their vulnerability windows. The open-label design has no discontinuation-driven dropout and is therefore most robust to dropout overall, but it cannot identify the BR -versus- PB split and is the wrong design for the predictive-biomarker question. The OL+BDC design supplies identifiability of the split but introduces a discontinuation-phase vulnerability window where treatment-responders may drop out from symptom return. The traditional crossover design has two discontinuation windows (one per crossover) and is more exposed to symptom-improvement-driven dropout. The hybrid N-of-1 design has the largest number of identifying contrasts at risk and therefore the largest set of vulnerability windows. The trade-off between identifiability and dropout vulnerability is the central design choice this paper analyzes.

2.3 What the published N-of-1 literature does and does not do

[9]

273 **The methodological N-of-1 corpus addresses dropout gener-**
274 **ically but does not engage the design-by-dropout coupling**
275 **this paper centers on.**

- 276 • Lillie et al. and Vohra et al. CONSORT extension codify report-
277 ing standards but do not prescribe analytic methods for informa-
278 tive dropout in N-of-1 contexts.
- 279 • Systematic reviews document that dropout rates are reported
280 without specifying mechanism or analytic treatment.
- 281 • Dropout is one of the leading methodological weaknesses in N-of-1
282 reporting audits.

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285 The methodological N-of-1 corpus addresses dropout as a generic chal-
286 lenge but does not engage the design-by-dropout coupling this paper
287 centers on. We briefly survey what has been broached and what has
288 not.

289 [10]

290 **Existing N-of-1 reporting standards codify dropout disclo-**
291 **sure but do not prescribe analytic methods for informative**
292 **dropout.**

- 293 • Lillie et al. and the Vohra CONSORT extension set reporting
294 standards without addressing informative-dropout analysis.
- 295 • Systematic reviews find that substantial proportions of N-of-1
296 studies report rates without specifying mechanism or analytic
297 treatment.
- 298 • Dropout is one of the leading methodological weaknesses in N-of-1
299 reporting audits.

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302 Lillie et al.¹¹ and the CONSORT extension of Vohra et al.¹² codify re-
303 porting standards that include dropout reporting but do not prescribe
304 analytic methods for handling informative dropout specifically in N-of-
305 1 contexts. The systematic reviews of Stunnenberg et al.¹³ and Nate-
306 san Batley and colleagues¹⁴ document that substantial proportions of
307 published N-of-1 studies report dropout rates without specifying the
308 dropout mechanism or its analytic treatment, and that dropout is one
309 of the leading methodological weaknesses identified in N-of-1 reporting
310 audits.

311 [11]

312 **Hendrickson (2020) is the foundational reference; this paper**
313 **extends it along three axes.**

- The extension uses wider parameter ranges (carryover half-lives, effect sizes, dropout rates) than the original.
- It examines sensitivity to dropout-pattern misspecification at the analysis stage, which the original assumed away.
- It offers an explicit decomposition of the happy-accident randomization-path selection effect, scoped for path-stratified production testing.

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¹ introduce the first quantitative analysis of dropout- by-design coupling in the aggregated N-of-1 context, with a hazard- based dropout model whose intensity depends on cumulative symptom worsening since baseline. Their Figure 4A shows the four-design power matrix as a function of dropout pattern, and their Section 3.4 bias- quantification establishes that under symptom-worsening-driven dropout the OL+BDC design develops a substantial negative bias in the biomarker-treatment interaction coefficient ($p < 0.0001$). The present paper takes¹ as the foundational reference and extends it along three axes. First, we use wider parameter ranges (longer carryover half-lives, higher biomarker effect sizes, larger dropout rates) than the original analysis. Second, we examine sensitivity to dropout-pattern misspecification at the analysis stage:¹ assumed the analyst knows the dropout mechanism, but in practice the analyst may model dropout as MCAR while the truth is MNAR, or may model it as symptom-worsening-driven when it is improvement-driven, and the present paper quantifies the inferential cost of each misspecification. Third, we offer an explicit decomposition of the ‘happy accident’ randomization-path selection effect.¹ noted that informative dropout preferentially preserves the most-informative crossover blocks in OL+BDC and hybrid N-of-1 designs (because patients with inconclusive open-label response are more likely to continue to the later blocks); in the present prototype we offer a marginal account of this effect, and a path-stratified test is scoped for the production rerun.

[12]

The paper’s contribution is to import classical missing-data, joint-modeling, and selection-bias machinery into the N-of-1 design-comparative simulation context.

- Little-Rubin taxonomy, selection models, and pattern-mixture models are adapted to the N-of-1 setting.
- Hazard-coupled-to-longitudinal-trajectory joint-modeling underwrites the Hendrickson dropout model and its extension.
- None of this machinery has previously appeared in the N-of-1

357 literature in a design-comparative simulation study.

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360 The classical missing-data literature^{3,4,15} supplies the methodologi-
361 cal scaffolding (MAR/MNAR taxonomy, selection models, pattern-
362 mixture models, multiple imputation) that we adapt to the N-of-1 set-
363 ting. The informative-dropout and joint-modeling machinery^{16–18} sup-
364 plies the hazard- coupled-to-longitudinal-trajectory formulation that
365 underwrites the¹ dropout model and its extension here, and the lon-
366 gitudinal mixed-models foundations¹⁹ supply the analytic backbone
367 for the $\beta_{bm:D}$ inference under MAR- plausible regimes. The clinical-
368 trial dropout literature^{6–8} supplies the parallel-group analogue against
369 which the design-comparative results below should be read, and the
370 selection-bias literature²⁰ supplies the template for thinking about how
371 response-based exclusion or informative dropout damages internal and
372 external validity. None of this machinery has been imported into
373 the N-of-1 methodology literature in a design-comparative simulation
374 study, and that import is the present paper’s contribution.

375 2.4 What this paper contributes

376 [13]

377 **The paper makes four contributions across Sections 2**
378 **through 6.**

- 379 • Section 2 formalizes the informative-dropout model and places it
380 in the missing-data taxonomy.
- 381 • Section 3 characterizes each design family’s phase-specific vulner-
382 ability windows.
- 383 • Sections 4 and 5 conduct and report a Monte Carlo simulation
384 crossing four designs against five dropout patterns.

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387 We make four contributions, each addressed in a section below.

388 [14]

389 **Section 2 formalizes the hazard-based informative-dropout**
390 **model and establishes its MNAR character explicitly.**

- 391 • The Hendrickson hazard model is placed in the MCAR/MAR/MNAR
392 taxonomy and shown to be MNAR.
- 393 • Four canonical dropout patterns plus a no-dropout reference are
394 defined as cells in the (β_0, β_1) hazard-parameter space.
- 395 • The operational definitions fix the simulation’s data-generating
396 mechanism for the subsequent studies.

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399 In Section 2 we formalize the informative-dropout model for aggre-
400 gated N-of-1 trials. The hazard-based model of¹ is placed in the
401 missing-data taxonomy, its MNAR character is established explicitly,
402 and the four canonical dropout patterns (balanced, more-of-flat, more-
403 of-biased, high-dropout, plus a fifth no-dropout reference) are defined
404 operationally as parameter cells in the (β_0, β_1) space of the hazard.

405 [15]

406 **Section 3 identifies the phase-specific dropout-vulnerability**
407 **windows and at-risk within-participant contrasts for each de-**
408 **sign family.**

- 409 • Each of the four design families is characterized by its exposure
410 to each dropout pattern per phase.
- 411 • Within-participant contrasts that are at risk of removal under
412 each pattern are identified explicitly.
- 413 • The characterization motivates the design-by-dropout-pattern
414 structure of the simulation in Sections 4 and 5.

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417 In Section 3 we characterize the four design families' dropout vulner-
418 ability windows. For each of the open-label, OL+BDC, traditional
419 crossover, and hybrid N-of-1 designs we identify the phase-specific ex-
420 posure to each dropout pattern and the within-participant contrasts
421 that are at risk of being removed.

422 [16]

423 **Sections 4 and 5 report a Monte Carlo study crossing four**
424 **designs against five dropout patterns at three sample sizes**
425 **and three effect sizes.**

- 426 • Performance measures include power, Type I error, bias, empiri-
427 cal SE, and path-conditional power decomposition.
- 428 • Reporting follows the Morris (2019) ADEMP framework with
429 Monte Carlo standard errors (MCSE) alongside every estimate.
- 430 • The randomization-path decomposition quantifies the happy-
431 accident selection effect.

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434 In Sections 4 and 5 we conduct and report a Monte Carlo simulation
435 study that crosses the four design families against the five dropout
436 patterns at three total sample sizes $(N \in \{35, 70, 100\})$, allocated

437 across each design’s randomization paths) and three biomarker-effect
438 sizes ($\beta_{bm} \in \{0, 0.3, 0.6\}$). The simulation reports power, Type I er-
439 ror, bias of $\hat{\beta}_{bm:D}$, mean standard error, and the randomization-path-
440 conditional power decomposition that quantifies the ‘happy accident’
441 selection effect. Reporting follows the² ADEMP framework adopted
442 as the project-wide simulation-reporting standard.

443 [17]

444 **Section 6 synthesises a recommendation table mapping**
445 **dropout mechanism class to preferred N-of-1 design choice.**

- 446 • Three mechanism classes (balanced, worsening-driven,
447 improvement-driven) are mapped to preferred designs indexed
448 by anticipated dropout rate.
- 449 • The table is presented as practical guidance for trial designers
450 selecting among the four canonical N-of-1 families.
- 451 • Section 7 discusses implications for trial-design selection and pm-
452 simstats operating characteristics.

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455 In Section 6 we synthesise a design-by-dropout-pattern recommenda-
456 tion table that maps three classes of dropout mechanism (balanced,
457 worsening- driven, improvement-driven) to preferred design choices in-
458 dexed by the anticipated overall dropout rate. The recommendation
459 is presented as guidance for trial designers selecting among the four
460 canonical N-of-1 design families when a particular dropout mechanism
461 is anticipated.

462 [18]

463 **Section 7 discusses implications for trial-design selection and**
464 **pmsimstats operating characteristics.**

- 465 • The discussion addresses design selection for biomarker-
466 validation N-of-1 trials.
- 467 • Operating characteristics of the pmsimstats simulation program
468 are reviewed.

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471 Section 7 discusses the implications for trial-design selection in
472 biomarker-validation N-of-1 trials and for the operating characteris-
473 tics of the pmsimstats simulation program.

474 3 Informative-dropout model

475 [19]

The censoring hazard combines a time-only ignorable component with a symptom-trajectory-biased component, made absorbing by a monotone carry-forward pass.

- The per-timepoint censoring probability is the union of two independent draws: a flat component scaling with elapsed time and a biased component scaling with the convex-transformed symptom change since baseline.
- Two hazard parameters govern the split: β_0 sets the overall dropout fraction and β_1 the proportion of that fraction routed through the symptom-biased component, with the convexity exponent e_{b1} fixed at two.
- A second carry-forward pass renders dropout absorbing: once a participant is censored at a timepoint, all later timepoints for that participant are censored.

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We adopt the hazard-based informative-dropout model implemented in the pmsimstats R/censordata.R infrastructure, which operationalizes the mechanism of¹. For participant i at timepoint t , the marginal censoring probability is the union (logical OR) of two independent Bernoulli draws,

$$\Pr(\text{drop}_{it}) = 1 - (1 - p_{it}^{\text{flat}})(1 - p_{it}^{\text{bias}}),$$

where the flat component $p_{it}^{\text{flat}} \propto t$ depends on elapsed time alone and is calibrated to a marginal dropout fraction of $\beta_0(1 - \beta_1)$, and the biased component $p_{it}^{\text{bias}} \propto t(\Delta_{Sx}(t) + 100)^{e_{b1}}$ depends on the symptom change $\Delta_{Sx}(t)$ since baseline and is calibrated to $\beta_0\beta_1$. The additive shift of 100 keeps the convex argument positive across the realised range of symptom scores, so that a large improvement lowers the dropout probability while a small change or a worsening raises it. A second pass enforces monotone dropout: once a participant is censored at a timepoint, every later timepoint for that participant is also set missing. The hazard is thus governed by three quantities: the overall rate β_0 , the biased proportion β_1 , and the convexity exponent e_{b1} , held at two throughout.

[20]

Whenever the biased component is active the mechanism is missing-not-at-random, because the censoring indicator at a timepoint depends on the symptom value that the same indicator renders missing.

- The flat component depends only on elapsed time, a fully observed design variable, and is therefore ignorable with respect to

- the latent response.
- The biased component makes dropout at time t depend on $\Delta_{Sx}(t)$, and because that observation is itself censored at the triggering timepoint the mechanism is MNAR in the Little-Rubin sense.
 - The departure from ignorability is controlled by β_1 : at $\beta_1 = 0$ the mechanism reduces to the ignorable limit, and the MNAR departure increases monotonically with β_1 .

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The two components occupy different rungs of the Little-Rubin ladder. The flat component depends only on elapsed time, which is a fully observed design variable rather than a function of the response, so on its own it induces missingness that is ignorable for likelihood-based inference. The biased component is the consequential one. Because the censoring indicator for the observation at time t is a function of $\Delta_{Sx}(t)$, the symptom change realised at that same timepoint, and because the triggering observation is itself rendered missing in the carry-forward pass, the missingness of Y_{it} depends on the value of Y_{it} . This is the defining condition for missing-not-at-random data^{3,4}. The mechanism is therefore MNAR for every pattern with $\beta_1 > 0$, and the standard MMRM appeal to a MAR justification⁶ does not hold. The extent of the violation scales with β_1 , which makes the biased proportion, not the overall rate, the methodologically load-bearing parameter.

[21]

Five dropout patterns are fixed as cells in the (β_0, β_1) hazard-parameter space, spanning a no-dropout reference through balanced, flat-weighted, bias-weighted, and high-rate conditions.

- The no-dropout reference sets $\beta_0 = 0$ and supplies the full-data gold standard against which the bias estimands of Study 2 are measured.
- Balanced ($\beta_0 = 0.05, \beta_1 = 0.5$), more-of-flat ($\beta_1 = 0.2$), and more-of-biased ($\beta_1 = 0.8$) hold the overall rate fixed while varying informativeness from mostly-ignorable to strongly MNAR.
- High-dropout ($\beta_0 = 0.15, \beta_1 = 0.5$) raises the overall rate at the balanced split, isolating the rate axis from the informativeness axis.

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The five dropout patterns are defined operationally as cells in the (β_0, β_1) parameter space of the hazard, with $e_{b1} = 2$ held throughout. The no-dropout reference ($\beta_0 = 0$) retains the complete data and

furnishes the gold-standard estimate of $\beta_{bm:D}$ used in the second bias-reference mode of Study 2. The balanced pattern ($\beta_0 = 0.05, \beta_1 = 0.5$) routes half of a modest five-percent dropout fraction through the symptom-biased component. Holding β_0 fixed, the more-of-flat ($\beta_1 = 0.2$) and more-of-biased ($\beta_1 = 0.8$) patterns move the same overall rate toward the ignorable and the strongly MNAR ends of the informativeness axis respectively, so that any difference in operating characteristics between them isolates the effect of the mechanism from the effect of the rate. The high-dropout pattern ($\beta_0 = 0.15, \beta_1 = 0.5$) triples the overall rate at the balanced split, isolating the complementary rate axis. The sign convention of the biased component is worsening-driven, in that a worsening relative to baseline raises the dropout probability; the symptom-improvement-driven variant examined in the design-recommendation of Section 6 is obtained by reflecting the symptom-change argument, so that improvement rather than worsening raises the hazard.

4 The four design families and their vulnerability windows

[22]

The four design families are distinguished by their phase structure, which determines both the within-participant contrasts that identify the biomarker-treatment interaction and the windows in which informative dropout can remove them.

- Each design is a set of randomization paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors of `buildtrialdesign`.
- The estimand-defining contrasts are within-participant differences across phases; informative dropout is damaging in proportion to how directly it removes those contrasts rather than merely reducing sample size.
- The four designs are ordered by an increasing number of identifying contrasts at risk, and therefore by an increasing number of vulnerability windows, from open-label through the hybrid N-of-1 design.

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Each candidate design is represented as a set of randomization paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors that `buildtrialdesign` accepts. The biomarker-treatment interaction is identified by within-participant contrasts across those phases, so the consequence of informative

599 dropout for a given design is determined by which phase a dropout
600 removes and which contrast that phase carries. We take the four
601 designs in order of increasing number of identifying contrasts at risk.

602 [23]

603 **The open-label design has a single all-on-drug path, no dis-**
604 **continuation window, and is the most dropout-robust but**
605 **cannot identify the pharmacological-versus-placebo split.**

- 606 • The single path carries only an on-drug-versus-baseline contrast,
607 which is confounded with natural history and cannot separate
608 the *BR* and *PB* components.
- 609 • With no off-drug phase there is no discontinuation-driven vulner-
610 ability window; dropout reduces only the precision of the on-drug
611 slope.
- 612 • The design is the robustness benchmark but is the wrong design
613 for the predictive-biomarker question.

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616 The open-label design comprises a single path on which every time-
617 point is on-drug. The only within-participant contrast available is
618 the on-drug response against baseline, which is confounded with the
619 natural-history trajectory *TV* and cannot separate the pharmacologi-
620 cal response *BR* from the placebo response *PB*. Because there is no
621 off-drug phase, there is no discontinuation-driven vulnerability win-
622 dow, and informative dropout can do no more than reduce the pre-
623 cision with which the on-drug slope is estimated. The design is con-
624 sequently the most robust to dropout of the four and serves as the
625 robustness benchmark, but its inability to identify the *BR*-versus-*PB*
626 split makes it the wrong instrument for the predictive-biomarker ques-
627 tion that motivates the program.

628 [24]

629 **The open-label-plus-blinded-discontinuation design adds a**
630 **silent-placebo phase that identifies the pharmacological-**
631 **versus-placebo split but opens a discontinuation window**
632 **in which worsening-driven dropout removes exactly that**
633 **contrast.**

- 634 • The estimand-defining contrast is the on-drug-versus-post-
635 discontinuation response, which separates *BR* from *PB*.
- 636 • Treatment-responders whose symptoms return on the silently
637 switched placebo may drop, mistaking the return for trial
638 failure, which is a worsening-driven dropout located in the
639 discontinuation phase.

- Because the lost phase carries the identifying contrast, dropout here biases $\hat{\beta}_{bm:D}$ rather than merely inflating its variance.

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The open-label-plus-blinded-discontinuation design retains the open-label phase and appends a blinded discontinuation phase in which the active drug is silently replaced by placebo. The discontinuation phase supplies the identifiability that the bare open-label design lacks: the contrast between the on-drug response and the post-discontinuation response separates the pharmacological component BR from the placebo component PB . That same phase is the design's vulnerability window. A treatment-responder whose symptoms return on the silently switched placebo may interpret the return as trial failure and drop out, a worsening-driven dropout concentrated in the discontinuation phase. The damage is more than a loss of precision: because the removed phase carries the very contrast that defines the estimand, dropout here produces a systematic negative bias in $\hat{\beta}_{bm:D}$, the effect quantified in¹ Section 3.4 and reproduced in Study 2.

[25]

The traditional crossover design has two randomization paths and two transition windows, and is robust to worsening-driven dropout but exposed to improvement-driven dropout at each crossover.

- The drug-first and placebo-first paths each carry a drug-block-versus-placebo-block contrast that identifies the within-participant treatment effect.
- A responder who feels well on the drug block and worsens after crossing to placebo may drop, an improvement-driven dropout that removes the second block and one crossover contrast.
- The two transition windows make the design more exposed than the single-window OL+BDC under improvement-driven dropout, though comparably robust under worsening-driven dropout.

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The traditional crossover design has two randomization paths, one beginning on drug and crossing to placebo and the other the reverse, each composed of two blocks separated by a transition. The identifying contrasts are the drug-block-versus-placebo-block differences carried within each path. The design has two transition windows, one per crossover, and its exposure is asymmetric in the dropout mechanism. Under worsening-driven dropout it is comparatively robust, because worsening tends to occur on the placebo block and a dropout there still

leaves the drug-block observations that anchor the contrast. Under improvement-driven dropout it is exposed: a responder who feels well on the drug block and then worsens after crossing to placebo may drop, removing the second block and with it one of the two crossover contrasts. The two transition windows therefore make the crossover design more vulnerable than the single-window OL+BDC design when the operative mechanism is improvement-driven.

[26]

The hybrid N-of-1 design has the largest number of identifying contrasts at risk and the most vulnerability windows, but recovers power through the happy-accident selection effect that preferentially preserves the most informative crossover blocks.

- An open-label run-in followed by a randomized multi-block crossover gives four randomization paths and the widest set of vulnerability windows.
- Participants with a clear open-label response are resolved early, so those continuing to the later randomized blocks are enriched for inconclusive open-label response.
- Biased dropout therefore preferentially preserves the most informative crossover blocks, partially offsetting the raw power loss from the larger vulnerability surface.

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The hybrid N-of-1 design of¹ combines an open-label run-in, which establishes each participant's apparent response, with a subsequent randomized multi-block crossover. With two randomization points it admits four paths and the largest number of identifying contrasts at risk of the four families, and therefore the widest set of phase-specific vulnerability windows: the run-in phase, each crossover block, and each transition between them. On raw power this makes the hybrid design the most dropout-exposed. The exposure is partly offset, however, by a selection effect that¹ terms a 'happy accident'. Participants whose open-label response is unambiguous, whether clear responders or clear non-responders, are the ones most readily resolved and most prone to leave early; those who continue into the later randomized blocks are consequently enriched for inconclusive open-label response, which is precisely the subgroup whose crossover contrasts are most informative for the biomarker-treatment interaction. Biased dropout thus preferentially preserves the most informative crossover blocks, partially recovering the power that the larger vulnerability surface would otherwise forfeit. The magnitude of this recovery is the object of the randomization-path decomposition reported in Study 3.

5 Simulation design

[27]

The simulation follows ADEMP with power as primary estimand and four secondary estimands covering bias, empirical SE, path decomposition, and misspecification sensitivity.

- Primary estimand is power for the $H_0 : \beta_{bm:D} = 0$ interaction test in the post-dropout linear-mixed analysis.
- Bias is reported in two modes: relative to the calibrated true value and relative to the full-data gold standard.
- Secondary estimands include path-conditional power (Study 3) and dropout-pattern misspecification sensitivity (Study 4).

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The simulation program follows the ADEMP framework of². The primary estimand is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear-mixed analysis fitted to the post-dropout retained data. Secondary estimands are the bias of $\hat{\beta}_{bm:D}$ (reported in two reference modes: relative to the calibrated true value and relative to the gold-standard estimate from the full uncensored data), the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decomposition that quantifies the ‘happy accident’ selection effect (Study 3), and the analysis-method sensitivity to dropout-pattern misspecification (Study 4).

Notation for the moderation parameter. The data-generating process used here is the mean-moderation architecture, so the biomarker-moderation parameter is written β_{bm} , the dimensionless multiplier scaling the treatment effect by the standardized biomarker b_i . The companion architecture manuscript reserves c_{bm} for the Architecture-B parameter, which is a correlation. The reference value 0.45 denotes a matched moderation strength under either architecture, but the symbols are not interchangeable.

[28]

Every performance measure is reported with its Monte Carlo standard error, following the project-wide ADEMP-audit recommendation.

- Power and Type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- Bias and empirical SE of $\hat{\beta}_{bm:D}$ carry the standard MCSE of the within-replicate mean.
- Reporting MCSEs alongside every estimate is a project-wide ADEMP-audit requirement.

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767 Performance measures are reported with MCSEs alongside every es-
768 timate, addressing the project-wide ADEMP-audit recommendation.
769 Power and Type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$; bias and em-
770 pirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard MCSE of the within-
771 replicate mean.

772 [29]

773 **Four pre-registered studies fix the simulation scope before**
774 **production runs: power, bias, path decomposition, and mis-**
775 **specification sensitivity.**

- 776 • Study 1 (180 cells) reproduces and extends Hendrickson Figure
777 4A for power as a function of design and dropout pattern.
- 778 • Study 2 (120 cells) addresses two-mode bias quantification; Study
779 3 (12 cells) examines the happy-accident path decomposition.
- 780 • Study 4 (60 cells) evaluates sensitivity to dropout-pattern mis-
781 specification at the analysis stage.

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784 The full Aims, Data-generating mechanisms, Estimands, Methods,
785 and Performance measures are pre-registered at `analysis/scripts/informative-dropout-by-design/0`
786 which fixes four studies before production runs. Study 1 (180 cells)
787 covers power as a function of design and dropout pattern, reproducing
788 and extending¹ Figure 4A. Study 2 (120 cells) addresses two-mode
789 bias quantification. Study 3 (12 cells with path- conditional decom-
790 position) examines the ‘happy accident’ randomization-path selection
791 effect. Study 4 (60 cells) evaluates sensitivity to dropout-pattern
792 misspecification at the analysis stage.

793 (Forthcoming detail. Pmsimstats infrastructure with the existing `R/censordata.R`
794 hazard-based dropout module. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and
795 bias estimands; $n_{\text{reps}} = 5000$ for Type I error cells.)

796 6 Results

797 [30]

798 **The prototype ran 500 replicates per cell across 16 cells with**
799 **100 percent convergence and achieved dropout fractions close**
800 **to targets.**

- 801 • 8-core `parallel::mclapply` completed 8,000 fits with 100 per-
802 cent convergence.

- Cell MCSE on power runs 0.008–0.019, widest in the binding 0.85 regime; at matched N no cell is saturated.
- Achieved dropout fractions tracked targets: approximately 0.25 for MCAR_25 and biased_25, approximately 0.40 for biased_40.

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The simulation program was executed at 500 replicates per cell across the 16-cell design-by-dropout-pattern grid using 8-core parallel::mclapply. Convergence was 100 percent across all 8,000 fits in every cell. Per-replicate output is at analysis/data/quick-sim/09-dropout-replicates.rds; the Morris ADEMP cell-level summary at analysis/data/quick-sim/09-dropout-summary.txt. Cell MCSE on power runs 0.008 to 0.019, widest in the binding 0.85 regime. At the matched total no cell is saturated, which matters for the loss comparisons below: the earlier unmatched run put the hybrid design at 140 participants and its power within a fraction of a point of 1.000, where a design cannot exhibit a loss whatever its structural merits. Achieved dropout fractions tracked targets closely: approximately 0.25 for both MCAR_25 and biased_25, approximately 0.40 for biased_40.

6.1 Power across designs and dropout patterns

[31]

At a matched total of $N = 70$ the Hendrickson happy-accident ordering holds, but the hybrid margin over OL+BDC is small.

- No dropout: hybrid 0.968 > OL+BDC 0.950 > traditional crossover 0.854 > OL 0.088.
- The ordering is the same at every dropout pattern, so it is a property of the designs and not of one dropout condition.
- The hybrid-over-OL+BDC gap reaches significance only under 25 percent uninformative dropout (5.4 points, $z = 3.59$); at no dropout (1.8, $z = 1.44$) and at 40 percent biased dropout (1.6, $z = 0.85$) the two are not separated.

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We begin with the power results. At the matched total of $N = 70$ the four designs reproduce the¹ happy-accident randomization-path ordering: without dropout the hybrid design (0.968) is followed by OL+BDC (0.950), then traditional crossover (0.854), then OL (0.088). That ordering is unchanged at all four dropout patterns, so it reflects the designs rather than a particular dropout condition, and because

every design carries the same total it is not a restatement of sample size. The ranking is nonetheless coarser than it looks at the top. The hybrid advantage over OL+BDC is 1.8 percentage points without dropout ($z = 1.44$), 2.4 under 25 percent worsening-driven dropout ($z = 1.55$) and 1.6 under 40 percent worsening-driven dropout ($z = 0.85$), none of them separable at 500 replicates per cell. It is separable in exactly one condition, 25 percent uninformative dropout, where the gap widens to 5.4 points ($z = 3.59$) because OL+BDC gives up 3.8 points there while the hybrid design gives up 0.2. Both designs are clearly separated from crossover throughout. What this establishes is that the branching after week 8 produces within-subject and across-path contrasts in D_{bc} that place the hybrid design in the top pair on equal sample, with a margin over OL+BDC that this run resolves only under uninformative dropout. Cell-level power values, percent $\pm$ MCSE:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	8.8 ± 1.3	9.4 ± 1.3	8.6 ± 1.3	11.2 ± 1.4
OL+BDC	95.0 ± 1.0	91.2 ± 1.3	92.4 ± 1.2	89.4 ± 1.4
traditional crossover	85.4 ± 1.6	81.0 ± 1.8	82.4 ± 1.7	76.8 ± 1.9
hybrid	96.8 ± 0.8	96.6 ± 0.8	94.8 ± 1.0	91.0 ± 1.3

[32]

The OL row carries almost no interaction information, which is a structural property of the design under the mean-moderation architecture rather than a dropout effect.

- Under the mean-moderation architecture, D_{bc} is identically one across all OL timepoints, so no within-subject $bm:D_{bc}$ contrast exists.
- Rejection rates sit at 0.086–0.112 across the four dropout cells, an order of magnitude below the other designs and flat with respect to dropout.
- They are nonetheless about twice nominal alpha, so the row is not a clean null: with the interaction term aliased, the reported rate is not a valid test size and should be read as an information floor, not as either power or Type I error.

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The OL row stays flat under every dropout pattern because the mean-moderation architecture's level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_{bc} identically one, there is no within-subject contrast for $bm:D_{bc}$ and the fitter has essentially no information about the interaction. This is a structural property of the OL design under the mean-moderation architecture, not a dropout arti-

fact, which is why the four OL cells vary only between 0.086 and 0.112 while every other design loses several points to heavy dropout.

Two cautions on how to read that row. It is not a power estimate, and it is also not the clean null the earlier unmatched run made it look like. At 35 participants the OL cells fell in 0.022 to 0.046, close enough to $\alpha = 0.05$ to invite the reading that the design returns exactly the nominal rate; at the matched 70 they sit near twice nominal. Because the interaction term is aliased with the biomarker main effect in this design, the quantity being reported is not a valid test size, and its numerical agreement with α at one sample size was a coincidence rather than a null-calibration check. The row is best read as an information floor: the level below which a design with no off-drug contrast cannot rise.

[33]

At matched N the hybrid and OL+BDC designs are equally vulnerable to informative dropout; only the crossover design is clearly worse.

- Absolute power loss from no-dropout to **biased_40**: hybrid 5.8 pp, OL+BDC 5.6 pp, traditional crossover 8.6 pp.
- Hybrid and OL+BDC are indistinguishable on this measure; the crossover loss is about half again as large.
- The hybrid design's apparent near-immunity in the earlier unmatched run (0.4 pp) was an artifact of its power sitting against the ceiling at twice the sample, not a structural property.

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Resilience to informative dropout, measured as absolute power loss from the no-dropout cell to the **biased_40** cell, is 5.8 percentage points for the hybrid design, 5.6 for OL+BDC and 8.6 for traditional crossover. The first two are the same to within Monte Carlo error; the crossover design loses about half again as much. So of the three powered designs the crossover is distinctly the most vulnerable, and the other two cannot be separated on this evidence.

This corrects a claim the earlier unmatched run supported and this one does not. At 140 participants the hybrid design lost 0.4 percentage points, which read as near-immunity to dropout and, taken with its 0.998 baseline, suggested the attractive conclusion that the most powerful design is also the most robust. Both halves of that were artifacts of sample size: a design whose power sits within a fraction of a point of 1.000 has almost no room to register a loss, so the small loss was a ceiling effect rather than evidence about structure. At matched N the hybrid design's baseline falls to 0.968 and its loss rises to 5.8 points, in

line with OL+BDC. The alignment between power and robustness is therefore not a finding of this study, and the practical design-selection advice that rested on it does not follow.

6.2 Bias of the biomarker-treatment interaction estimate

[34]

The mean estimated interaction coefficient is stable across dropout cells within each design, indicating no systematic bias in the estimand.

- $\hat{\beta}_{bm:D_{bc}}$ shows no systematic trend across none, MCAR_25, biased_25, and biased_40 cells within any design.
- Stability holds for the three powered designs (OL+BDC, crossover, hybrid); the OL design is uninformative on bias.
- The absence of bias in the estimand implies the power degradation traces to standard-error inflation, not bias.

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We turn next to the bias of the biomarker-treatment interaction estimate. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells within each design, indicating no systematic bias in the estimand under informative dropout in this regime:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	-\$0.09	-\$0.10	-\$0.09	-\$0.10
OL+BDC	-\$2.48	-\$2.48	-\$2.47	-\$2.45
traditional crossover	-\$2.24	-\$2.28	-\$2.25	-\$2.21
hybrid	-\$2.46	-\$2.46	-\$2.46	-\$2.45

[35]

Power degradation under informative dropout traces to standard-error inflation, not bias in the estimand: empirical SEs expand modestly under biased_40 dropout.

- The largest within-design bias shift (approximately 0.04 for traditional crossover) is within cell-level MCSE.
- Empirical SEs expand under biased_40: OL+BDC 0.628 to 0.660, crossover 0.766 to 0.779, hybrid 0.411 to 0.454.
- The power loss in Section 5.1 is therefore attributable to SE inflation rather than systematic bias.

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The largest within-design bias shift between the no-dropout and the `biased_40` cells is approximately 0.04 (traditional crossover), well within the cell-level MCSE on the mean (≤ 0.04 for all designs). The power degradation visible in Section 5.1 therefore traces to inflation of the standard error of $\hat{\beta}_{bm:D_{bc}}$, not to systematic bias in the estimand. Empirical SEs (last column of `09-dropout-summary.txt`) expand modestly under `biased_40` dropout: from 0.628 to 0.660 for OL+BDC; from 0.766 to 0.779 for traditional crossover; and from 0.411 to 0.454 for hybrid.

6.3 Randomization-path decomposition

[36]

Path-stratified decomposition of the happy-accident effect is scoped for a production rerun; the current prototype provides only marginal results consistent with the hypothesis.

- The current prototype aggregates across paths within each design and cannot attribute robustness to specific surviving paths.
- The Hendrickson happy-accident hypothesis is consistent with the marginal results but requires path-tracked data for a formal test.
- The follow-up extension requires recording `path_id` per replicate and adding path-stratified $\hat{\beta}_{bm:D_{bc}}$ summary logic.

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We note that decomposition of the headline power loss into randomization-path-specific contributions is scoped for a production-grade rerun with per-path-id tracking added to the driver. The current prototype aggregates across paths within each design and therefore cannot directly attribute the hybrid design's robustness to specific paths surviving informative dropout. The substantive¹ 'happy accident' hypothesis (that informative dropout preferentially preserves the most-informative crossover blocks) is consistent with the present marginal results, but a formal test against the alternative (dropout uniform across paths) requires the path-tracked data that the current prototype does not produce. The follow-up driver extension is relatively straightforward at the simulation level, requiring only that `path_id` be recorded per replicate; the bottleneck is in the analysis layer, where path-stratified $\hat{\beta}_{bm:D_{bc}}$ distributions need to be summarized against the cell-pooled estimate.

6.4 Sensitivity to dropout-pattern misspecification

[37]

994 **The biased_25 versus MCAR_25 contrast isolates the bias mech-**
995 **anism from the rate effect; within-design differences are small**
996 **and statistically indistinguishable at 500 replicates per cell.**

- 997 • OL+BDC: biased_25 minus MCAR_25 equals +1.2 pp (CI
998 [−2.4, +4.8]); crossover: +1.4 pp (CI [−3.5, +6.3]).
- 999 • Hybrid: −1.8 pp; OL: −0.8 pp on aliased cells; all differences lie
1000 within or near 1.96 cell-MCSE.
- 1001 • Resolving the distinguishability question would require approxi-
1002 mately 2,000 replicates per cell at the current MCSE.

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1005 The biased_25 versus MCAR_25 contrast at fixed 25 percent dropout
1006 isolates the bias mechanism from the rate effect. The within-design
1007 differences are small in absolute terms: OL+BDC biased_25 power
1008 minus MCAR_25 power equals +1.2 percentage points (CI [−2.4, +4.8]);
1009 traditional crossover equals +1.4 percentage points (CI [−3.5, +6.3]);
1010 hybrid equals −1.8 percentage points; and OL equals −0.8 percentage
1011 points on cells where the interaction term is aliased and the rate is not
1012 a valid test size. None of the four is distinguishable from zero at 500
1013 replicates per cell.

1014 [38]

1015 **The dominant power determinant is dropout rate rather than**
1016 **bias mechanism; mechanism effects are not statistically dis-**
1017 **tinguishable at 500 replicates per cell.**

- 1018 • Moving from 25% to 40% biased dropout costs 3.0 pp (OL+BDC),
1019 5.6 pp (crossover), 3.8 pp (hybrid).
- 1020 • The crossover rate-effect loss is clearly detectable above MCSE;
1021 the OL+BDC and hybrid losses are borderline.
- 1022 • Resolving the biased_25 versus MCAR_25 mechanism question
1023 requires approximately 2,000 replicates per cell, deferred to a
1024 production rerun.

1025 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*
1026 *tempor incididunt ut labore et dolore magna aliqua.*

1027 All four within-design differences lie within 1.96 cell-MCSE and
1028 are not statistically distinguishable from zero at this sample size.
1029 The dominant effect on power is the dropout rate rather than the
1030 bias mechanism: moving from 25 percent to 40 percent biased
1031 dropout costs 3.0 percentage points (OL+BDC), 5.6 percentage
1032 points (traditional crossover), and 0.6 percentage points (hybrid),
1033 with the crossover loss clearly detectable above MCSE. We note that
1034 resolving the biased_25 versus MCAR_25 distinguishability question

with statistical clarity would require approximately 2,000 replicates per cell at the current MCSE, which we leave to a production rerun.

7 Design-by-dropout recommendation

[39]

Three classes of anticipated dropout mechanism are mapped to preferred N-of-1 design choices based on the Section 5 simulation results.

- The mapping is indexed by anticipated overall dropout rate within each mechanism class.
- The recommendation draws directly on the power and bias results from the 16-cell prototype simulation.
- The table is intended as practical guidance for trial designers selecting among the four canonical N-of-1 families.

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In this section we shall map three classes of anticipated dropout mechanism (no informative dropout, informative-by-rate, and informative-by-mechanism) to preferred N-of-1 design choices, drawing on the Section 5 results.

Anticipated dropout	Preferred design	Rationale
None or low MCAR ($d \leq 10\%$)	hybrid or OL+BDC	0.968 versus 0.950 at $N = 70$; the gap is not separable ($z = 1.44$), so complexity is the deciding factor
Moderate MCAR ($d \approx 25\%$)	hybrid	Hybrid 0.966, OL+BDC 0.912, crossover 0.810; the only condition separating the top pair ($z = 3.59$)
Heavy MCAR or biased ($d \geq 40\%$)	hybrid or OL+BDC	Hybrid 0.910, OL+BDC 0.894, crossover 0.768; crossover is the only clear loser
Worsening-driven (any rate)	hybrid or OL+BDC	OL+BDC's blinded discontinuation phase carries the bias signal explicitly
Improvement-driven (any rate)	OL+BDC	Crossover loses signal when responders drop early; OL+BDC absorbs
Information-poor (OL only)	not recommended	OL under mean moderation has no within-subject $bm:D_{bc}$ contrast

[40]

The dominant recommendation message is that the hybrid and OL+BDC designs are interchangeable on these criteria

and the traditional crossover is the one to avoid under heavy dropout.

- Hybrid and OL+BDC are not separable on power at matched N except under 25 percent uninformative dropout, and are indistinguishable on dropout loss, so the choice between them mostly turns on cost and complexity.
- Traditional crossover loses approximately 9 percentage points under `biased_40` and is the least dropout-robust powered design.
- The OL design lacks the within-subject D_{bc} contrast under the mean-moderation architecture and does not warrant inclusion in the primary recommendation.

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The dominant message from the table is narrower than the earlier unmatched run suggested. On equal sample the hybrid Hendrickson design and OL+BDC are not separable on absolute power in three of the four dropout conditions, and are indistinguishable on power lost to dropout, so the table recommends the pair rather than the hybrid alone in those rows and the choice between them turns on burden and feasibility. The exception is 25 percent uninformative dropout, the one condition where the hybrid design's margin is resolvable, and the table recommends it there. The traditional crossover, despite respectable no-dropout power, loses approximately 9 percentage points under `biased_40` dropout and is the least dropout-robust of the three powered designs, which is the clearest discrimination the study supports. The OL design lacks the within-subject D_{bc} contrast under the mean-moderation architecture and is uninformative about the interaction at any dropout level; it therefore does not warrant inclusion in the primary recommendation.

8 Discussion

8.1 Prototype Monte Carlo confirmation

[41]

The 500-replicate prototype confirms 100 percent convergence across all 8,000 fits with achieved dropout fractions close to targets.

- Executed at 35 participants per randomization path ($N = 35$, 70, 70, 140 in total for OL, OL+BDC, crossover and hybrid), $\beta_{bm} = 0.45$, the mean-moderation architecture on 8 cores in 421 s with 100 percent convergence.
- Achieved dropout: approximately 0.25 for `MCAR_25` and

1097 `biased_25`; approximately 0.40 for `biased_40`.

1098 • Cell MCSE on power runs 0.008–0.019; at matched N no cell is

1099 saturated.

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1102 A 500-replicate-per-cell prototype run on the full 16-cell design-

1103 by-dropout grid (design $\in \{\text{OL}, \text{OL+BDC}, \text{traditional crossover},$

1104 $\text{hybrid}\} \times \text{dropout pattern} \in \{\text{none}, \text{MCAR 25\%}, \text{biased 25\%}, \text{biased}$

1105 $40\%\}$) was executed at $\beta_{bm} = 0.45$ under the mean-moderation

1106 architecture using 8-core `parallel::mclapply`. Wall clock was 421 s;

1107 convergence was 100 percent across all 8,000 fits.

1108 Every design runs at the same total, $N = 70$, allocated across that

1109 design’s randomization paths: 70 on the single path of OL, 35 on each

1110 path of OL+BDC and traditional crossover, and 18/18/17/17 across

1111 the four paths of the hybrid design. The design comparison is therefore

1112 matched on N .

1113 An earlier version of this run fixed the per-path count at 35 instead,

1114 which gave the four designs totals of 35, 70, 70 and 140 and made

1115 the design ordering inseparable from a 1:2:2:4 sample-size ratio. The

1116 figures below are from the matched re-run. Per-replicate output is

1117 at `analysis/data/quick-sim/09-dropout-replicates.rds`; the

1118 Morris ADEMP summary is at `09-dropout-summary.txt`. Achieved

1119 dropout fractions tracked targets closely: approximately 0.25 for both

1120 `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`. Cell

1121 MCSE on power runs 0.008–0.019, widest in the binding 0.85 regime;

1122 at the matched total no cell is saturated.

1123 [42]

1124 **The prototype reproduces the Hendrickson happy-accident**

1125 **power ordering at matched N : `hybrid` > `OL+BDC` >**

1126 **`crossover` > `OL`.**

1127 • Power ordering without dropout: hybrid 0.968, OL+BDC 0.950,

1128 crossover 0.854, OL 0.088.

1129 • The branching after week 8 gives the hybrid design within-subject

1130 and across-path D_{bc} contrasts, placing it in the top pair on equal

1131 sample.

1132 • The ordering holds across all four dropout-pattern cells, though

1133 the hybrid-versus-OL+BDC gap is resolvable only under 25 per-

1134 cent uninformative dropout.

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1137 The four designs reproduce the¹ happy-accident randomization-path

ordering at the wider grid: without dropout, hybrid (0.968) is followed by OL+BDC (0.950), then traditional crossover (0.854), then OL (0.088, on cells where the interaction term is aliased). The hybrid four-path design places at the top because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} , though on equal sample its margin over OL+BDC is resolvable only under 25 percent uninformative dropout.

This ordering is produced at a common total of $N = 70$, so it is a statement about the designs rather than about their sample sizes. An earlier version of this run gave the hybrid design 140 participants against 70 for the two-path designs, which made its advantage inseparable from a doubled sample; the matched re-run removes that confound, so whatever margin the hybrid design holds here is attributable to its branching structure rather than to its size.

[43]

The OL design’s nominal-alpha rejection rate is a structural property of the mean-moderation architecture, not an artifact of dropout, and serves as an implicit null check.

- Under the mean-moderation architecture, $D_{bc} \equiv 1$ for all OL timepoints, eliminating any within-subject $bm:D_{bc}$ contrast.
- The fitter has no interaction information, so rejection rates collapse to nominal alpha across all dropout conditions.
- This confirms the OL row is a structural null check rather than evidence of dropout robustness.

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The OL row sits at nominal alpha under every dropout pattern because the mean-moderation architecture’s level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_{bc} identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal Type I rate. This is a structural property of the OL design under the mean-moderation architecture, not a dropout artifact; the OL row therefore functions as an implicit null check, not a power estimate.

[44]

Dropout resilience is dominated by the rate effect, not the bias mechanism; the linear-mixed model with corCAR1 is relatively unbiased under informative dropout in this regime.

- Absolute power loss from no-dropout to `biased_40`: hybrid 0.004, OL+BDC 0.056, crossover 0.086.
- The `biased_25` vs. `MCAR_25` mechanism contrast is within or

1179 barely outside 1.96 MCSE; the rate increase from 25 to 40 percent
1180 is the dominant effect.

- 1181 • $\hat{\beta}_{bm:D_{bc}}$ is stable (−2.45 to −2.48), confirming power degradation
1182 traces to SE inflation rather than bias in the estimand.

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1185 Resilience to informative dropout, ranked by absolute power loss from
1186 the no-dropout cell to the `biased_40` cell, runs as follows: hybrid
1187 (loss approximately 0.004) above OL+BDC (loss approximately
1188 0.056) above traditional crossover (loss approximately 0.086). The
1189 `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout is
1190 small (+0.012, +0.014, −0.038 for OL+BDC, crossover, and hybrid
1191 respectively), all within or barely outside 1.96 MCSE. The dominant
1192 effect is dropout rate, not bias mechanism: moving from 25 percent
1193 to 40 percent biased dropout costs 0.006–0.056 power across designs,
1194 while doubling the bias proportion at constant rate is essentially
1195 noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$
1196 is stable across dropout cells (−2.45 to −2.48 for OL+BDC and
1197 hybrid), suggesting that the linear-mixed model with `corCAR1` is
1198 relatively unbiased under informative dropout in this regime; the
1199 power degradation traces to standard-error inflation rather than
1200 systematic bias in the estimand.

1201 [45]

1202 **Two ADEMP caveats limit the prototype’s scope: no formal**
1203 **null cells for non-OL designs, and single-architecture cover-**
1204 **age.**

- 1205 • No $\beta_{bm} = 0$ cells were run for OL+BDC, crossover, or hybrid;
1206 production reruns should add explicit null cells for each.
- 1207 • The prototype is single-architecture (the mean-moderation archi-
1208 tecture); under the covariance-moderation architecture the OL
1209 row carries information via the covariance structure.
- 1210 • Resolving the `biased_25` vs. `MCAR_25` distinguishability question
1211 requires approximately 2,000 replicates per cell.

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1214 Two ADEMP-discipline caveats warrant mention. First, no formal
1215 Type I error cells under $\beta_{bm} = 0$ were run for the non-OL designs;
1216 the OL row serves as a structural null check by argument, not as a
1217 designed null cell. A canonical production run should add explicit
1218 $\beta_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the
1219 prototype is single-architecture (the mean-moderation architecture)
1220 by specification; under the covariance-moderation architecture (multi-

variate normal, MVN) the OL row would carry information because the biomarker-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the `biased_25` versus MCAR_25 distinguishability question would, as noted above, require approximately 2,000 replicates per cell at the current MCSE.

9 Conclusions

[46]

The conclusions distil three findings: design-by-dropout differentiation, rate dominance over mechanism, and SE inflation as the power-loss mechanism.

- The four N-of-1 design families differ materially in dropout robustness in ways the published literature has not characterized.
- Dropout rate dominates mechanism as the determinant of power loss; mechanism becomes load-bearing only when rates are comparable.
- Estimand stability under informative dropout is reassuring but also a warning: coverage degrades as SEs widen without the model acknowledging it.

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In conclusion, three points merit emphasis.

[47]

First conclusion: the traditional crossover is the least dropout-robust of the powered designs, while the hybrid and OL+BDC designs are equally robust at matched sample size.

- At a matched total of $N = 70$, $\beta_{bm} = 0.45$, the mean-moderation architecture: hybrid loses 5.8 pp, OL+BDC 5.6 pp and crossover 8.6 pp under `biased_40`.
- OL lacks the within-subject D_{bc} contrast under the mean-moderation architecture and is not viable for the biomarker-interaction question.
- The hybrid design's apparent near-immunity in an earlier unmatched run was a ceiling effect at twice the sample, so the happy-accident argument survives as a power claim but not as a robustness claim.

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First, the four candidate aggregated N-of-1 trial designs differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterized. At the prazosin/PTSD reference parameters with a matched total of $N = 70$ for every design, $\beta_{bm} = 0.45$, and the mean-moderation architecture, the traditional crossover is the most vulnerable of the three powered designs, losing approximately 8.6 percentage points of power under 40 percent biased dropout. The hybrid Hendrickson design and OL+BDC lose 5.8 and 5.6 percentage points respectively, which is the same to within Monte Carlo error, so this study separates the crossover from the other two but does not rank those two against each other. The OL design lacks the within-subject D_{bc} contrast required for the biomarker-treatment interaction under the mean-moderation architecture.

An earlier unmatched version of this analysis reported the hybrid design losing less than half a percentage point, and concluded that the most powerful design was also the most robust. That conclusion does not survive matching on N : at 140 participants the hybrid design's power sat within a fraction of a point of 1.000, where almost no loss can be registered, and at 70 its loss is in line with OL+BDC. The¹ happy-accident randomization-path argument is therefore corroborated here as a statement about power, where the hybrid design places in the top pair on equal sample, but not as a statement about differential robustness to dropout.

[48]

Second conclusion: dropout rate is the dominant power determinant; mechanism is load-bearing only when rates are comparable across competing design candidates.

- The `biased_25` vs. `MCAR_25` contrast at fixed 25 percent dropout is below MCSE detection at 500 replicates.
- Moving to 40 percent biased dropout produces losses in all three powered designs, clearest in the crossover.
- Trial designers should evaluate designs against anticipated dropout rate first, and mechanism only secondarily.

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Second, the dominant determinant of the design-by-dropout interaction, in this prototype's evaluation, is the dropout rate rather than the dropout mechanism. The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout produces within-design differences below MCSE detection at 500 replicates per cell; moving to 40 percent biased dropout, by contrast, produces clear power losses in every design except the hybrid. Trial designers should therefore evaluate proposed N-of-1 de-

signs against the anticipated dropout rate in their target population first, and only secondarily against the bias mechanism. The mechanism becomes the load-bearing factor in the regime where the rate is comparable across competing trial-population candidates.

[49]

Third conclusion: estimand stability is reassuring but also a coverage warning, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

- The largest within-design shift in $\hat{\beta}_{bm:D_{bc}}$ is below 0.04 across the no-dropout to biased_40 range.
- Power degradation traces to SE inflation; the inferential target is preserved under informative dropout in this regime.
- Nominal CI coverage may understate true uncertainty: the coefficient distribution widens but the model does not acknowledge it.

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Third, the mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells, with the largest within-design shift below 0.04 across the no-dropout to biased_40 range. The power degradation under informative dropout therefore traces to standard-error inflation rather than systematic bias in the estimand. This finding is reassuring for the validity of the aggregated N-of-1 framework in the presence of informative dropout, in the sense that the inferential target is preserved, but it is also a warning to trialists that nominal coverage of confidence intervals may understate the true uncertainty in the dropout regime, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

[50]

Three production-grade extensions are scoped: path tracking, the covariance-moderation architecture coverage, and higher replicate counts to resolve the biased-versus-MCAR distinguishability question.

- Per-path-id tracking is needed for the randomization-path decomposition of the happy-accident effect.
- The covariance-moderation architecture coverage is needed to interpret the OL row as a power estimate rather than a structural null check.
- Resolving the biased_25 vs. MCAR_25 question requires 2,000 or more replicates per cell.

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Production-grade extensions of the prototype (per-path tracking for the randomization-path decomposition, the covariance-moderation architecture for the OL design’s null-check elevation, and 2,000 or more replicates per cell to resolve the biased-versus-MCAR distinguishability question) are scoped in Section 5 of this manuscript and represent the natural follow-up to the present proof-of-concept evidence.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project’s `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The informative-dropout module is at `R/censordata.R`.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/09-dropout-replicates.rds`. Morris ADEMP cell-level summaries are at the companion `09-dropout-summary.txt`. Driver scripts are under `analysis/scripts/informative-dropout-by-design/` and `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. The pre-registered ADEMP plan is at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registra`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/09-informative-dropout-by-design/`.

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